

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Short Answer Test

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)*

Assessment Tool Weightage: 40%

Dave's Taxonomy: Imitation (Level 1)
Question Count: 5

Total Marks: 25

Code of Conduct

I NS avesh hussain (192525241) _____(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action. Signature of the student: NS avesh hussain__

Assessment

Tasks Short

Answer Test

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.
2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.
3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.
4. Discuss the role of **data centers** in delivering cloud services..
5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

Short Answer Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected sub-points completely	Covers most expected points	Covers only basic points	Very limited coverage
Use of Examples	15%	Uses highly relevant cloud examples to strengthen the answer	Uses relevant examples with minor mismatch	Uses vague or partially relevant examples	No useful examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow

CO1 - Assessment Tool-1 - Short Answer test

- i) Cloud computing: Cloud computing is a technology that provides computing such as server storage data and analysis, Data Base Management and make them reliable and superintendent of the structure and computing a physical storing device server space can be bought and hold the fitting together federations and stores the data in the data centers cloud computing is a pay what we use service we only pay for what amount of storage we use.

Characteristics:

- * Cloud computing is more efficient because we only pay for what amount of storage space we use the space can be easily increased or decreased based on demand.
- * Uploading and retrieving data is more efficient now this saves the time
- * Data sharing is done online many system's can be hosted with single server
- * Rapid elasticity based on customer's demand.

2) Evolution of cloud computing:

The evolution of cloud computing from hardware evolution to internet software and the changes are have been listed below.

i) Main frame era:

In 1960, a Big Main screen will be there where everyone will go work with a Big Powerful screen. These kind of screens are used by business persons in those time. This system will be big and cost high and also requires main frame.

ii) Personal Computers era (1980's)

The time Personal Computers can every person willing to have a system can own one with storage hardware's.

iii) Networking and internet:

Introduction of internet made data sharing more easy and efficient without hardware we can share data and also obtain data from websites.

iv) virtualization:

one server can be divided into multiple because virtual machine systems. This is the most efficient way because of this process.

3) server virtualization

server virtualization is the process of dividing one physical server into multiple virtual machine where each VM works like an independent computer with its own OS and application.

It improves hardware utilization by allowing multiple workloads to run on a single server reducing the need for additional physical hardware.

virtualization helps reduce cost saves energy simplifies server management and supports faster development application.

virtualization is enabling tech that creates virtual resources such as servers storage and network it forms the foundation of cloud computing.

Differences between both of them?

virtualization focuses on creative and moving the virtualization and it helps to make it more reliable to speed up the compilation time and by that the execution time will be increase rapidly.

4)

Role of Data centers in delivering cloud services:

Data centers hold the data to share with fundamental parts of the unique resolved substances into the hardware and it separates the work flow and the data of the storage managing devices for the manipulation of its own contained works and can make changes in servers to generate the modules for the delivering data to the correct conceptual storing devices.

Role of Data centers in ~~the~~ Cloud Computing:

→ Data centers help users to store data for cloud computing without the term process will be published of because the main theme of cloud computing is to store the extended data & use it effectively so without any data cloud computing can't run.

Data centers play a main role in cloud computing.

5) IaaS: (Infra structure as a service):-

→ This service depends on the infrastructure such as machines, web browsers etc.

ex: AWS EC2

PaaS: (Platform as a service)

→ This service depends on platforms such as software, websites, apps etc.

ex: Google, One.

* SaaS (Software as a service)

This service gives everything to use software, and Provider manages everything and user can use the software.

ex: Gmail.

* XaaS: (Everything as a service):

→ we can use everything as a service in this service we can use software (or) Platform at the number of sites to come manage